Second Year Research Proposal

Ke Jiang

Project Title: Local Control and Metropolitan Inefficiency: The Interaction between Zoning and Transportation Investment

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1 Introduction

1.1 Motivation

Transportation infrastructure is a critical component of urban economic activity. It improves access to jobs and amenities and can shape the spatial structure of cities (Redding, 2025). However, the returns to transportation infrastructure depend critically on local conditions, such as housing supply restrictions that limit density and ridership responses to new infrastructure (Tsivanidis, 2026; Severen, 2021). Existing literature often treats housing supply constraints as given, but in practice these constraints are frequently the result of local zoning policies. Such policies are locally determined and politically supported by incumbent residents, who value neighborhood amenities and property values (Rollet and Weiwu, 2025). Although restrictive zoning may generate local benefits, it can also impose spillover costs on the broader metropolitan area and distort transportation infrasructure. This proposal therefore asks: how does local zoning distort metropolitan transportation investment decisions, what is the optimal transportation network when zoning and transportation policies are coordinated, and what are the welfare implications?

This proposal focuses in particular on transit investment decisions for two reasons. First, unlike road networks, whose benefits accrue to car commuters distributed broadly across the metropolitan network, transit investment generates returns concentrated at fixed station nodes. These returns depend critically on whether residents and employers can locate close enough to stations (Hu and Wang, 2025). Roads, by contrast, serve car commuters across a diffuse network of routes and do not require density to be concentrated at any particular location. This node-based return structure makes transit investment especially sensitive to local zoning. Second, empirical evidence suggests that the United States has lagged behind other countries in transit infrastructure development. Freemark, 2023 reports that U.S. transit expansion appears to have stagnated since the 1980s (fig. 1). I hypothesize that the asymmetric response of transit and road investment to local zoning may be an important factor contributing to this stagnation.

Underinvestment in transit may also generate a self-reinforcing dynamic through residential sorting. When transit is undersupplied, households substitute toward car-based commuting, which expands the feasible commuting radius and allows households to sort into more distant, lower-density locations (Baum-Snow, 2007). As population disperses into previously undeveloped areas, newly incorporated municipalities may emerge whose residents have selected into low-density living and therefore have strong political incentives to adopt restrictive zoning that preserves the neighborhood character they moved there to enjoy (Mast, 2024). This creates new constituencies for restrictive zoning in areas that might otherwise have remained undeveloped, further lowering metropolitan density and the ridership base for future transit expansion. In this sense, the mechanism studied in this proposal may produce not only a static inefficiency but also a path-dependent process through which metropolitan areas continue to sprawl. This process has important policy implications. Low-density sprawl requires more road infrastructure per capita, and construction and maintenance costs may become unsustainable (Brueckner, 2000). Car dependence and long commuting distances can also increase congestion costs that are not internalized by households. Moreover, low transit investment can limit low-income households' access to job opportunities, exacerbating economic inequality (Glaeser, Kahn, and Rappaport, 2008).

To answer the questions raised above, I plan to develop a quantitative spatial model with endogenous zoning regulation and transportation investment, following Fajgelbaum and Schaal, 2020. The model will be calibrated to U.S. metropolitan data and used to evaluate both the existing transportation network, especially the transit network, and the counterfactual optimal network under coordinated zoning and transportation policies. I will also evaluate welfare differences between the two scenarios to estimate the potential welfare gains and identify the winners and losers from policy coordination.

1.2 Literature Review

This proposal draws on three interconnected bodies of literature: studies of transportation infrastructure and its evaluation, literature on zoning and land-use regulation, and research on urban sprawl and city growth patterns.

1.2.1 Transportation Infrastructure, Optimal Structure, and Evaluation

A large body of literature examines how transportation infrastructure shapes spatial economic patterns and how to evaluate infrastructure investments. Donaldson (2025) provides a comprehensive framework for evaluating transportation infrastructure impacts, emphasizing the importance of general equilibrium effects and showing how infrastructure changes affect economic activity through multiple channels. The quantitative spatial economics framework is essential for this evaluation: Redding (2025) surveys quantitative urban economics methods, while Redding and Rossi-Hansberg (2017) details the development of quantitative spatial models that can integrate multiple dimensions of infrastructure impacts. Contemporary applications of these methods highlight that traditional benefit-cost analyses may not capture full infrastructure impacts. Severen (2026) describes how quantitative spatial models can evaluate transportation improvements within cities, providing practical guidance for implementation. Importantly, Fajgelbaum, Khandelwal, et al. (2021) develops a framework integrating spatial epidemiology and commuting networks, demonstrating the importance of accounting for spatial structure in policy design. The returns to transit infrastructure specifically have received attention. Tsivanidis (2026) evaluates Bogotá's TransMilenio system and finds that transit returns depend critically on complementary policies affecting land use. Baum-Snow and Kahn (2000) studies rail transit expansion in five U.S. cities and finds modest impacts on usage and housing values, suggesting limitations to transit investment absent other changes. Kahn (2007) documents significant heterogeneity in transit effects across 14 cities, with gentrification outcomes varying substantially based on local conditions like station type (walk-and-ride versus park-and-ride).

The interaction between transportation infrastructure and congestion is increasingly recognized. Akbar et al. (2023) analyzes the relationship between mobility and congestion in urban India, showing how congestion affects the effectiveness of transportation improvements. Allen and Arkolakis (2022) develops a general equilibrium framework with endogenous congestion to evaluate welfare impacts of transportation improvements, finding substantial variation in returns across different infrastructure investments. Kashner and Ross (2025) relates congestion to upzoning, showing that the effect of upzoning on housing supply depends on where upzoning occurs. Specifically, upzoning near transit stations has lower congestion effects than upzoning in other areas. My proposal contributes by modeling the endogeneity of zoning regulations and transportation investment, which are often taken as given in the literature.

1.2.2 Zoning and Land-Use Regulation

Zoning and land-use regulations have emerged as critical determinants of urban form and economic efficiency. The foundational work of Glaeser and Gyourko (2002) demonstrates that zoning restrictions significantly reduce housing supply elasticity, with particularly severe constraints in high-demand cities. This finding has profound implications: Hsieh and Moretti (2019) quantifies the aggregate costs of housing constraints, estimating that zoning restrictions lowered U.S. aggregate growth by 36 percent between 1964 and 2009 by preventing the spatial reallocation of labor to high-productivity locations.

Recent work has continued this examination with new evidence and methods. Rollet and Weiwu (2025) measures winners and losers from increasing housing supply, highlighting the distributional tensions inherent in zoning reform—local residents benefit from restrictive zoning even when it imposes broader costs. Song (2025) estimates the effects of residential zoning on U.S. housing markets, providing detailed evidence of how zoning shapes housing supply responses. Mast (2024) uses electoral reforms as a source of exogenous variation in local control and finds that decentralization of land-use authority reduces housing permits by 20 percent, with larger effects in affluent areas. These results highlight how local political economy affects housing supply. Importantly, Baum-Snow and Duranton (2025) surveys housing supply and affordability, providing context for understanding how zoning interacts with other factors affecting housing markets. A critical insight from recent work is that land-use regulations interact with transportation investment. Hu and Wang (2025) studies how subway expansion affects outcomes under different zoning regimes, finding that rigid land-use regulation limits the benefits of transit infrastructure. My proposal seeks to endogenize zoning regulation and investigate its spillover effects on aggregate transportation investment decisions.

1.2.3 Urban Sprawl and City Growth

Urban sprawl represents a fundamental concern in metropolitan economics. Brueckner (2000) provides the canonical analysis of sprawl causes and remedies, identifying three market failures: incomplete internalization of open-space benefits, failure to charge commuters for congestion externalities, and insufficient cost-recovery for infrastructure. Baum-Snow (2007), discussed above, documents empirically how highways have driven suburban decentralization, with implications for sprawl patterns. Glaeser and Kahn (2003) documents that sprawl is ubiquitous and continuing to expand, but is not caused by bad urban planning; rather, it reflects increasing demand for car-based living, though low-income people who cannot afford cars are left behind. My proposal seeks to explain another mechanism through which car ownership may be not only a preference but also an equilibrium outcome.

The literature collectively demonstrates that sprawl outcomes reflect a complex interaction between fundamental forces, such as population growth, rising incomes, falling transport costs, and structural changes. Cuberes, Desmet, and Rappaport (2021) documents urban growth shadows and access effects, showing how proximity to large metropolitan areas affects growth patterns through both competition and market-access mechanisms. Desmet and Rappaport (2017) traces the long-term evolution of U.S. settlement patterns from 1800-2000, documenting the transition toward increasingly dispersed city systems. Structural change and land use are intimately connected. Coeurdacier, Oswald, and Teignier (2022) studies how structural change in the economy drives land-use changes and urban expansion. International evidence adds important insights: Zheng and Kahn

(2013) documents how China's bullet trains facilitate market integration and mitigate megacity growth costs by allowing workers to access large labor markets from secondary cities. Building on these integrated insights, my proposal explores a potential mechanism that may further contribute to urban sprawl.

2 Methodology

2.1 Planned Empirical Work

The empirical part of the project will examine whether metropolitan transportation investment is shaped by the expected economic returns to transit. I plan to focus on three related hypotheses:

1. Transportation investment decisions respond to the expected costs and benefits of new infrastructure.
2. Older cities that developed extensive transit systems before widespread automobile dominance have higher returns to transit investment and therefore maintain larger transit networks, whereas newer, more car-oriented cities face lower returns.
3. Restrictive zoning lowers the return to transit investment by limiting the density and ridership response around high-access locations.

The main data sources will include the National Transit Database (NTD) and the Transit Explorer database, which provide information on transit volumes, revenues, investment, and opening dates. I will measure zoning constraints using housing supply elasticity estimates from Baum-Snow and Han (2024).

2.2 Quantitative Spatial Model

Following Severen (2026), I aim to develop a quantitative spatial model to answer the questions raised in the introduction. The model will capture the key features of metropolitan spatial structure, including heterogeneous locations, commuting patterns, and land-use decisions. It will also incorporate the institutional structure of local zoning and metropolitan transportation investment.

For now, I am considering a model in which local municipalities choose zoning policies to maximize incumbent welfare, while a metropolitan transportation authority chooses transportation investment to maximize regional net benefits, taking zoning as given. The key distortion arises from the fact that municipalities do not internalize the metropolitan-wide effects of their zoning decisions on housing supply, commuting patterns, and the return to transit investment.

2.2.1 Setup

Consider a metropolitan area consisting of a finite set of locations $S = \{1, 2, \ldots, N\}$. These locations are partitioned into a set of municipalities $g \in G$, such that

$$S = \bigcup_{g \in G} S^g, \qquad S^g \cap S^{g'} = \emptyset \quad \text{for } g \neq g'.$$

Each location $i \in S$ is endowed with a fixed amount of land $\bar{H}_i$, which can be allocated between residential and commercial use:

$$H_i^R + H_i^F \leq \bar{H}_i.$$

Land use is further subject to zoning regulations. Let Z_i^R and Z_i^F denote the residential and commercial zoning restrictions at location i. These regulations impose upper bounds on feasible development:

$$H_i^R \leq \bar{H}_i^R(Z_i), \qquad H_i^F \leq \bar{H}_i^F(Z_i),$$

where stricter zoning lowers the feasible density of development.

Locations are connected through a transportation network. For any pair of residence and workplace locations (i, j), households can choose a commuting route $r \in R_{ij}$. Following Fajgelbaum and Schaal, 2020, the generalized commuting cost associated with route r is denoted by τ_{ijr}, which depends on transportation infrastructure and congestion:

$$\tau_{ijr} = \tau_{ijr}(I_{ij}, \mathbf{n}),$$

where I_{ij} denotes the transportation investment between locations i and j, and $\mathbf{n}$ collects equilibrium traffic and transit usage across routes, which generate congestion. The commuting cost is increasing in the quantity of commuters on the route

$$\frac{\partial \tau_{ijr}}{\partial n_{ijr}} \geq 0, \tag{1}$$

and I_{ij} leads to a reduction in the commuting cost

$$\frac{\partial \tau_{ijr}}{\partial I_{ij}} \leq 0. \tag{2}$$

Following Fajgelbaum and Schaal, 2020, I will parametrize the commuting cost function as follows:

$$\tau_{jk}(\mathbf{n}, I) = \delta_{jk}^{\tau} \frac{(\mathbf{n})^{\rho}}{\mathbf{I}^{\gamma}} \tag{3}$$

There are three types of agents in the model: households, firms, and governments. Households choose where to live, where to work, and how to commute. Competitive firms choose labor and commercial land to produce a freely tradable good. Government decisions are made at two levels. Municipal governments choose local zoning regulations, while a metropolitan transportation authority chooses transportation investment for the urban area as a whole.

2.2.2 Households

A household chooses a residence location i, a workplace location j, and a commuting route $r \in R_{ij}$. Conditional on these choices, the household consumes a composite tradable good Q_{ij} and residential land h_{ij}^R. Preferences are given by

$$u_{ijr}(w) = \frac{B_i}{\tau_{ijr}} \left(\frac{Q_{ij}}{\beta}\right)^{\beta} \left(\frac{h_{ij}^R}{1-\beta}\right)^{1-\beta} v_{ijr}(w),$$

where B_i is the residential amenity at location i, $\beta \in (0, 1)$, and $v_{ijr}(w)$ is an idiosyncratic preference shock.

Let q_i^R denote the rental price of residential land at location i, and w_j the wage earned at workplace location j. The household's budget constraint is

$$Q_{ij} + q_i^R h_{ij}^R = w_j.$$

Solving the within-choice consumption problem yields the indirect utility

$$V_{ijr}(w) = \frac{B_i}{\tau_{ijr}} \frac{w_j}{(q_i^R)^{1-\beta}} v_{ijr}(w),$$

up to a constant common to all choices.

2.2.3 Firms

At each workplace location j, competitive firms produce a freely tradable good using effective labor and commercial land. Let output be given by

$$Y_j = A_j F(L_j, H_j^F),$$

where A_j is location-specific productivity, L_j is labor employed at location j, and H_j^F is commercial land use. Firms solve

$$\max_{L_j, H_j^F} \; A_j F(L_j, H_j^F) - w_j L_j - q_j^F H_j^F,$$

where q_j^F denotes the rental price of commercial land.

Profit maximization implies standard labor-demand and land-demand conditions. In equilibrium, wages and commercial rents adjust so that labor and land markets clear.

2.2.4 Municipal Zoning Decisions

Each municipality $g \in G$ chooses a vector of zoning policies

$$Z^g = \{Z_i\}_{i \in S^g}.$$

To avoid modeling an explicit dynamic voting game, I assume municipal governments act as reduced-form representatives of incumbent residents. This captures the idea that local land-use policy is shaped by the interests of existing residents, who value neighborhood amenities, lower congestion, and higher property values, but do not internalize metropolitan-wide effects on housing supply, commuting, or transit efficiency.

Accordingly, municipality g solves

$$\max_{Z^g} \; W^g = \sum_{i \in S^g} \omega_i u_i(Z^g, Z^{-g})$$

2.2.5 Optimal Transportation Investment

A metropolitan transportation authority chooses investment for the region as a whole. Taking local zoning as given, the authority solves

$$\max_{I_{jk}} \; W = \sum_i \omega_i \sum_j N_{ij} U_{ij}(Q_{ij}, h_{ij}^R)$$

subject to:

$$\sum_{j,k} \delta_{jk} I_{jk} \leq K, \tag{4}$$

$$0 \leq \underline{I_{jk}} \leq I_{jk} \leq \bar{I_{jk}} \leq \infty \tag{5}$$

Where $U_{ij}(Q_{ij}, h_{ij}^R)$ represents the utility of a household living in location i and working in location j, and N_{ij} is the number of such households.

2.2.6 Equilibrium

A decentralized equilibrium consists of

$$\left\{q_i^R, q_i^F, w_j, H_i^R, H_i^F, L_j, Z^g, I^R, I^T\right\}_{i,j \in S,\, g \in G}$$

such that:

1. Households choose residence, workplace, and commuting route to maximize utility.
2. Firms choose labor and commercial land to maximize profits.
3. Each municipality chooses zoning to maximize incumbent resident welfare.
4. The metropolitan transportation authority chooses road and transit investment to maximize regional net benefits, taking zoning as given.
5. Residential land, commercial land, and labor markets clear.

2.2.7 Inefficiency and the Source of Sprawl

The main distortion in the model arises from a mismatch in governance scales. Zoning is chosen locally by municipalities that represent incumbent interests, while transportation is chosen at the metropolitan level. Municipalities have incentives to restrict development in high-access locations in order to preserve local amenities and limit congestion. However, they do not internalize the broader metropolitan costs of such restrictions, including higher housing prices, greater job-housing separation, and lower returns to public transit investment.

As a result, restrictive zoning can prevent transportation improvements—especially transit investments—from translating into higher-density development near well-connected locations. Growth is instead pushed toward lower-cost, more distant areas, contributing to urban sprawl. In this sense, sprawl in the model is not simply a reflection of geography or household preferences, but may emerge as an inefficient equilibrium outcome of fragmented land-use regulation combined with metropolitan transportation investment.

2.2.8 Efficient Benchmark: Planner's Problem

Now, for a planner who chooses zoning and transportation investment jointly to maximize metropolitan welfare, the problem can be written as

$$\max_{Z^g, I_{jk}} W = \sum_i \omega_i \sum_j N_{ij} U_{ij}(Q_{ij}, h_{ij}^R)$$

subject to:

$$\sum_{j,k} \delta_{jk} I_{jk} \leq K, \tag{6}$$

$$0 \leq \underline{I_{jk}} \leq I_{jk} \leq \bar{I_{jk}} \leq \infty, \tag{7}$$

$$\sum_j N_{ij} h_{ij}^R \leq H_i^R, \tag{8}$$

$$\sum_i \sum_j N_{ij} = L, \tag{9}$$

$$\sum_i \sum_j N_{ij} Q_{ij} = \sum_j A_j F(L_j, H_j^F), \tag{10}$$

$$H_i^R + H_i^F \leq \bar{H}_i \tag{11}$$

3 Expected Outcome

I expect to use my model to quantify the existing U.S. metropolitan transportation network and construct the counterfactual optimal transportation network under coordinated zoning and transportation policies. I will also evaluate the welfare distribution and differences between the two scenarios to estimate how much welfare improvement can be achieved and identify the winners and losers.

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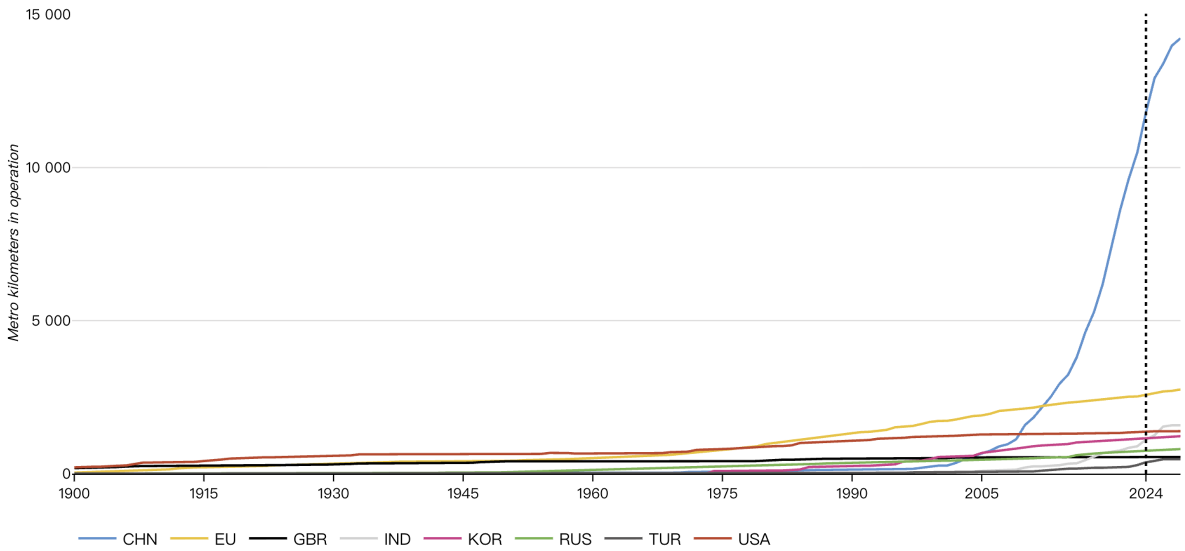

Figure 1: Metro kilometers in operation

4 Work Plan

The project runs from May 13 to August 3, 2026, spanning twelve weeks and approximately 57 working days (excluding Memorial Day, May 25, and Independence Day, July 4). Work is organized in three overlapping phases. **Phase 1** (Weeks 1–5) covers model specification and coding; the model structure determines what data series are needed, so data collection begins only once the model is sufficiently specified. **Phase 2** (Weeks 3–10) covers data collection, cleaning, and model calibration; data work and model coding run in parallel during Weeks 3–5 and feed into calibration from Week 7 onward. **Phase 3** (Weeks 10–12) covers counterfactual simulations and writing. Model development and data work may iterate if calibration reveals gaps in data coverage or model specification.

Activities

Table 1: Activities and Time Commitment

#	**Activity**	**Weeks**	**Working Days**
1	Model specification and theoretical derivations	W1–W3	13
2	Data requirements specification and collection	W3–W5	18
3	Data cleaning and descriptive analysis	W6–W8	12
4	Model coding and numerical solution algorithm	W3–W5	13
5	Model calibration and validation	W7–W10	16
6	Counterfactual simulations and welfare analysis	W10–W11	10
7	Writing and revisions	W10–W12	12
	Total (accounting for overlap)	**12 weeks**	**≈57 days**

Note: Activities 2 and 4 run in parallel during Weeks 3–5. Activity 5 draws on both the coded model (Activity 4) and cleaned data (Activity 3), and therefore begins only after both are sufficiently advanced.

Milestones

Schedule

Table 2: Project Milestones

M#	Date	Day	Deliverable
M1	May 30	13	Model fully specified; equilibrium conditions and key equations derived; data requirements identified
M2	June 13	23	Model coded and numerically solved; all datasets collected
M3	July 3	37	Data cleaned; descriptive statistics and stylized facts documented
M4	July 18	47	Model calibrated to U.S. metropolitan data; fit validated
M5	July 25	52	Counterfactual simulations and welfare calculations complete
M6	Aug 3	57	Summer paper draft submitted to faculty advisor

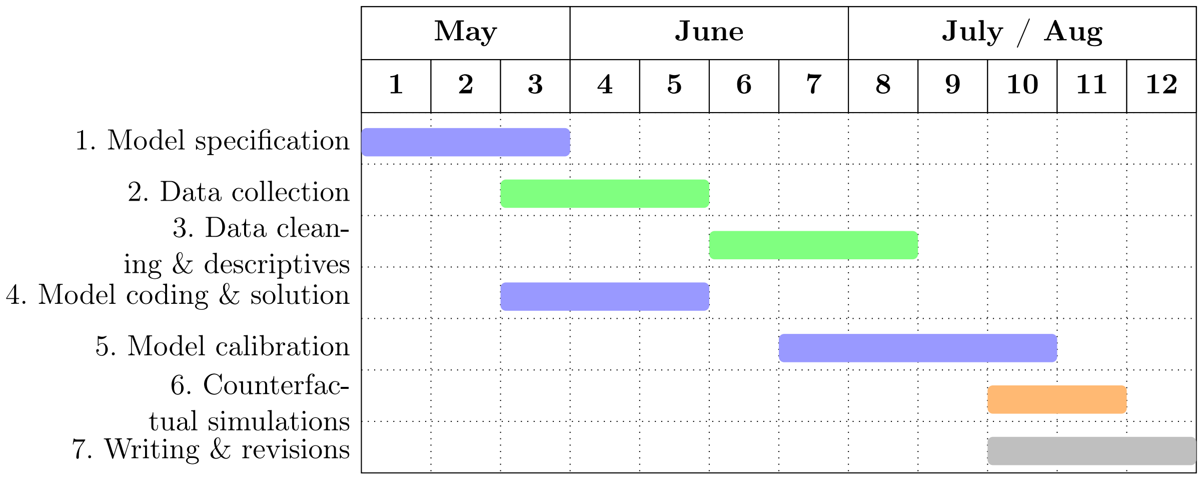

Figure 2: Summer work plan, May 13–August 3, 2026. Week numbers run from Week 1 (May 13) to Week 12 (July 28–August 1); the submission deadline of August 3 is the first working day of Week 13. *Blue*: model activities (1, 4, 5); *green*: data activities (2, 3); *orange*: simulation (6); *gray*: writing (7). Activities 2 and 4 occupy the same weeks (W3–W5) and run in parallel.